

PhD in Machine Learning specialized in interpretability to build safe, reliable, and trustworthy AI for critical applications.

Education

- 2025 **Dr. rer. nat. (summa cum laude)** **Technical University Berlin** Advisor: Prof. Dr. Wojciech Samek
Thesis: *Interpretability-Driven Model Debugging with the Reveal2Revise Life Cycle*
- 2018 **M.Sc. in Information Systems** **Humboldt University Berlin** Grade: 1.2 (very good), top 10%
- 2017 **Semester abroad** **University of Oslo** Master level courses
- 2015 **B.Sc. in Information Systems** **DHBW Mannheim** Grade: 1.4 (very good), top 10%

Work Experience

- 2021 – today **Research Associate** **Fraunhofer Heinrich-Hertz Institute** Berlin (Germany)
PhD student in the eXplainable AI group led by Dr. Lapuschkin and Prof. Dr. Samek until completing the PhD in 2025, now postdoctoral researcher in the same group after a 6-month sabbatical. Focus on interpretability-driven model debugging and, more recently, the interpretability of multi-agent systems. Development and integration of XAI methods into the AI development life cycle to verify, validate, and improve model behavior, ultimately increasing the reliability and trustworthiness of AI systems, particularly in medical applications. Technical lead for Fraunhofer in the EU Horizon 2020 project iToBoS, which developed a full-body scanner for AI-based early detection of skin cancer.
- 2019 – 2021 **Machine Learning Scientist** **Amazon.com, Inc.** Edinburgh (United Kingdom)
Machine Learning Scientist focusing on explainability and fairness in ML for recruiting technologies. Contributed to production systems and data pipelines, and led the development of a candidate-role matching prototype that was later launched.
- 2017 – 2019 **Master's Thesis Student / Research Associate** **SAP AI Research** Berlin (Germany)
Master's thesis student and full-time research associate with a focus on multimodal few-shot learning.
- 2017 **Machine Learning Developer (Intern)** **Bakken & Bæck** Oslo (Norway)
Text embedding learning, language detection, data retrieval, data preparation, and prototype development.
- 2015 – 2017 **Working Student** **SAP SE – Medical Research Insights** Potsdam (Germany)
Infrastructural topics, product development and ML prototypes for medical research insights.
- 2012 – 2015 **Bachelor Student (Cooperative Education)** **SAP SE** Walldorf (Germany)
During my bachelor's program I was employed by SAP SE and have worked in several development teams.

Selected Publications

*equal contribution · full list on Google Scholar

- NeurIPSW** **Safety Signals to Verify NetOps Agents with Action-Level Granularity**
2026 T. Labarta, **F. Pahde**, N. Boškov, M. Dreyer, D. Birkenberger, M. A. Khan, S. Lapuschkin, W. Samek · to appear
- ICLR** **Navigating Neural Space: Revisiting Concept Activation Vectors to Overcome Directional Divergence**
2025 **F. Pahde**, M. Dreyer, L. Weber, M. Weckbecker, C. Anders, T. Wiegand, W. Samek, S. Lapuschkin
- Journal** **Ensuring Medical AI Safety: Interpretability-Driven Detection and Mitigation of Spurious Model Behavior and Associated Data**
2025 **F. Pahde**, T. Wiegand, S. Lapuschkin, W. Samek · Machine Learning, Special Issue on XAI for Secure Applications
- AAAI** **From Hope to Safety: Unlearning Biases of Deep Models via Gradient Penalization in Latent Space**
2024 M. Dreyer*, **F. Pahde***, C. Anders, W. Samek, S. Lapuschkin · oral
- MICCAI** **Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models**
2023 **F. Pahde***, M. Dreyer*, W. Samek, S. Lapuschkin
- CVPRW** **Optimizing Explanations by Network Canonization and Hyperparameter Search**
2023 **F. Pahde**, G. Yolcu, A. Binder, W. Samek, S. Lapuschkin
- WACV** **Multimodal Prototypical Networks for Few-shot Learning**
2021 **F. Pahde**, M. Puszcz, T. Klein, M. Nabi

Academic Service

Reviewer ICLR, NeurIPS, TMLR, World Conference on Explainable AI, CVPR Workshops

Computer Skills

Programming Python, PyTorch, PyTorch Lightning, TensorFlow, Keras, scikit-learn, zennit, Java, R
Other Tools Git, Docker, Slurm, Weights & Biases, $\LaTeX$

Languages

German Native

English Fully Professional